CSE3038 – Spring 2021- Quiz 1 (35 mins)

1. Assume the following instruction mix

ALU	Load/Store	Branch
20%	30%	50%

Assume the CPI values of the instruction types on two machines are given below:

Machine	Clock Speed	ALU	Load/Store	Branch
A	4.5 Ghz	3	X	3
B	1.5 Ghz	1	1	3

What should be the CPI of Load/store for machine A (i.e. the value of X) so that machine A is two times as fast as machine B.

2. Assume that the variables k, m, are assigned to registers $t1, $t2, respectively. Assume that the base address of the arrays A and B are in registers $s1 and $s2, respectively. Translate the following C code to MIPS code.

k = A[2+B[m]];

Question 1

$$\frac{\text{Performance}_A}{\text{Performance}_B} = \frac{\frac{1}{\text{CPUtime A}}}{\frac{1}{\text{CPUtime B}}} = \frac{\text{CPUtime B}}{\text{CPUtime A}} = 2$$

$$\text{CPUtime}_B = \frac{IC \times CPI}{\text{Clock rate}} = \frac{I(1*0.2 + 1*0.3 + 3*0.5)}{1.5}$$

$$= \frac{2I}{1.5}$$

$$\text{CPUtime}_A = \frac{I(3*0.2 + Y*0.3 + 3\times 0.5)}{4.5}$$

$$\text{CPUtime}_A = \frac{I(2.1 + 0.3Y)}{4.5}$$

$$\frac{\text{CPUtime}_B}{\text{CPUtime}_A} = \frac{2I/1.5}{(2.1+0.3Y)I/4.5} = 2$$

$$\frac{2I}{1.5} \times \frac{4.5}{(2.1+0.3Y)I} = 2 \qquad \frac{6}{2.1+0.3Y} = 2 \qquad Y = 3$$

Question 2

k → $t1 — A (Base) → $s1

m → $t2 — B (Base) → $s2

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sll  $t0, $t2, 2      # $t0 = 4xm
add  $t0, $t0, $s2    # Addr. B[m]
lw   $s3, 0($t0)      # $s3 = B[m]
addi $s3, $s3, 2      # $s3 = B[m]+2
sll  $t0, $s3, 2
add  $t0, $t0, $s1    # Addr. A[2+B[m]]
lw   $t1, 0($t0)      # k = A[2+B[m]]
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